

Machine Learning I: Hands-On 7

7.1 Clustering Algorithms

Consider a dataset composed of instances described by two numerical attributes, A_1 and A_2 .

Point	A_1	A_2
p_1	4	4
p_2	0	2
p_3	4	3
p_4	0	0
p_5	5	4
p_6	1	1
p_7	2	0
p_8	5	5

Using the **Manhattan distance**, apply the following clustering algorithms to partition the data:

$$d_1(\mathbf{x}, \mathbf{z}) = \sum_j |x_j - z_j|.$$

1. Apply **K-means** with $k = 2$, using p_1 and p_6 as the initial centroids. Determine the final cluster assignments and the final centroids.
2. Apply **DBSCAN** with $\text{MinPts} = 2$ and $\varepsilon = 1$. Classify each point as **core**, **border**, or **noise**.
3. Apply **Hierarchical Agglomerative Clustering** with **single-link** merging. Show all merging steps and construct the corresponding dendrogram.

What to pay attention to

The three algorithms use different definitions of cluster structure. K-means relies on proximity to representatives, DBSCAN relies on local density, and single-link hierarchical clustering relies on pairwise connectivity. Compare not only the final clusters, but also *why* each algorithm creates them.